

PX Analytics Progress Report

MVP Architecture and Implementation

[Github](#)

Problem Statement

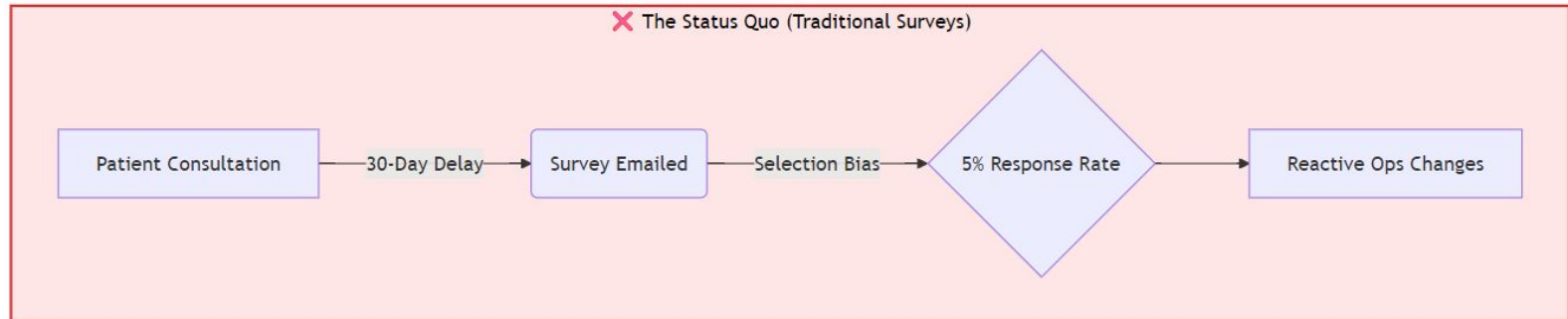
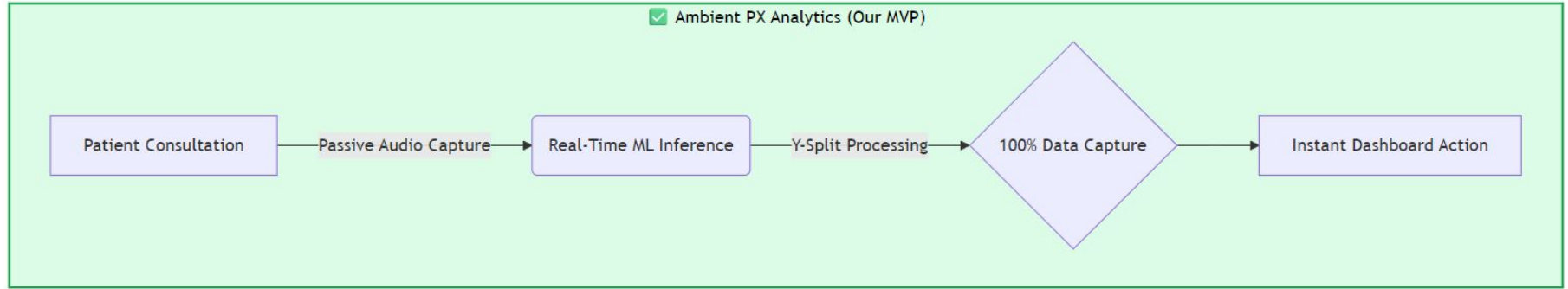
The Problem (Survey Bottleneck):

- Post-visit surveys capture a low ~5% response rate.
- Extreme selection bias (skewed toward highly satisfied or highly dissatisfied outliers).
- Reactive data lagged by 30 days, disconnected from clinical workflows.

The Solution (Ambient PX):

- Passive, ambient analysis of consultation audio ensuring a 100% capture rate.
- **Y-Split Processing:** Decouples acoustic biometrics (tone, tempo) from semantic clinical charting to preserve patient privacy.
- Actionable Output: Translates raw audio markers into standardized 0–100 metrics: Empathy, Clarity, and Efficiency.

Problem Statement

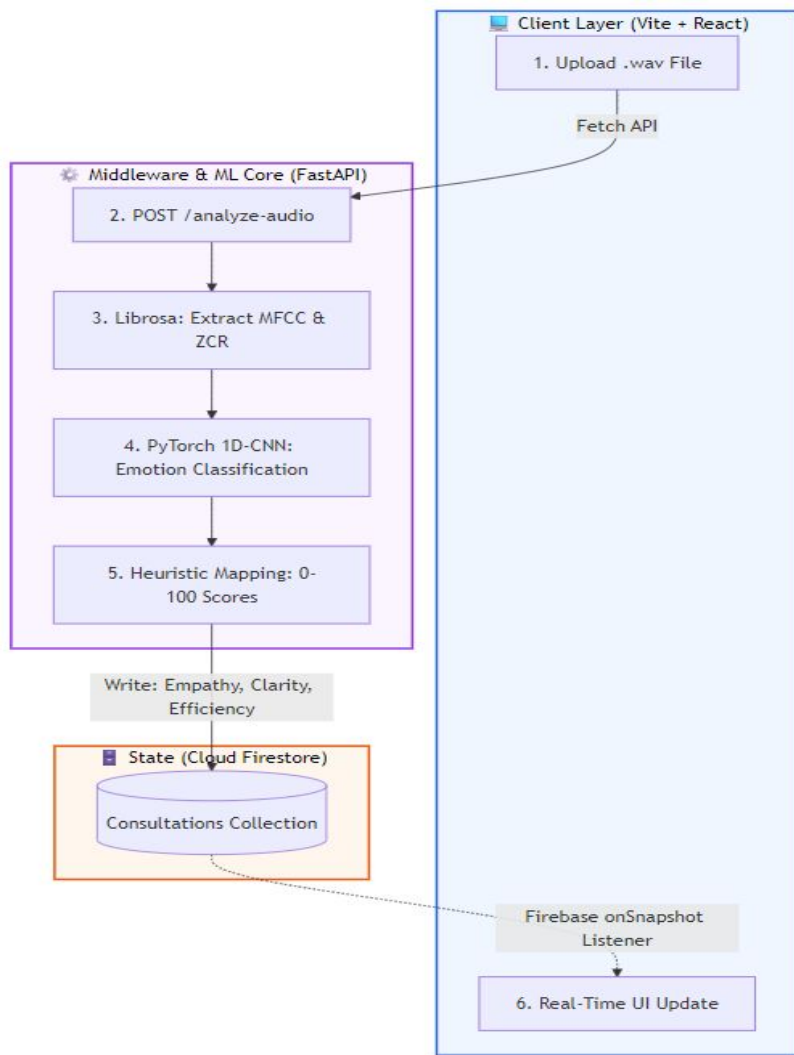


System Architecture & Tech Stack

- **Frontend Layer (UI & Orchestration):**
 - Vite, React 18
 - Tailwind CSS v4
 - Recharts
 - Lucide React.
- **Backend Layer (ML Inference API):**
 - Python 3.13, FastAPI, Uvicorn, PyTorch, Librosa, NumPy.
 - Lightweight 112,968 parameter 1D-CNN optimized for low-latency CPU inference.
- **Database & Security Layer (Data State):**
 - Cloud Firestore (NoSQL), Firebase Authentication.
 - Role-Based Access Control (RBAC) enforced strictly via server-side Firestore Security Rules.

End-to-End Operational Workflow

- 1) **Ingestion:** User uploads a .wav file via the Vite client interface.
- 2) **Inference:** File forwarded directly to the localized Python FastAPI server.
- 3) **Feature Extraction:** Librosa processes audio into a (41, 173) feature tensor (MFCCs + ZCR).
- 4) **Classification:** PyTorch 1D-CNN generates probability logits across 8 emotional classes.
- 5) **Heuristic Mapping:** Translation layer converts raw logits into 0–100 business scores.
- 6) **Live Storage & Sync:** Client appends the active provider_id, writes to Cloud Firestore, and triggers real-time Recharts visualizations via onSnapshot.



End-to-End Operational Workflow

Phase 1 — Acoustic Sentiment Engine

- **Objective:** Extract experiential markers without Natural Language Processing (NLP) or transcriptions to guarantee compliance.
- **Dataset & Training:** Trained on the RAVDESS emotional speech database (1,440 records, 8 distinct emotion classes).
- **Target Audio Features:**
 - 40 MFCCs: Captures vocal tract spectral shapes to isolate emotional tone
 - 1 Zero-Crossing Rate (ZCR): Tracks acoustic frequency transitions to evaluate speech tempo and articulation.
- **Mapping Logic Formulas:**
 - Empathy: Weighted positively by Calm/Happy logits; penalized by Angry/Fearful logits.
 - Clarity: Driven by a Gaussian curve centered on an ideal ZCR pacing coefficient (0.06).
 - Efficiency: Composite score balancing positive emotional valence against total stress markers.

Phase 2 & 3 — Infrastructure & Orchestration

- **API Integration:** Enabled CORS Middleware on the FastAPI server to permit secure client-side fetch streams directly from the Vite application domain.
- **Firestore Schema Design**
- users: Stores UID, display names, and designated authorization roles.
- providers: Tracks profiles, specializations, and historical performance rolling metrics.
- consultations: Houses immutable logs composed of timestamp, provider_id, patient_anxiety_flag, and the {metrics} map.
- **RBAC Constraints:**
- Providers: Restricted via security rules to read only their specific session radar charts and timelines.
- Admins: Unrestricted global read access across all hospital registries and benchmarking matrices.

Phase 4 — Live Dashboard Activation

- **Reactive State Plumbing:** Eradicated hardcoded mock JSON arrays; bound Recharts data attributes directly to live Firestore document snapshot data streams.
- **Data Seeding Script:** Implemented a standalone Node.js utility utilizing the Firebase Admin SDK to batch-inject 150 randomized consultations backdated across 30 days.
- **Simulated Visual Trends:** Programmed the seeder to weight scores higher as timestamps approached the current date, generating an organic, trending performance line for demo visualization.

Next Steps

- **Model Scale:** Transition from the lightweight 1D-CNN to a fine-tuned foundational audio Transformer architecture (e.g., Wav2Vec 2.0) for increased noise resilience.
- **Data Security Hardening:** Build automated ephemeral data processing loops to immediately purge raw .wav binaries post-inference to maximize privacy.
- **Bias Evaluation:** Implement continuous demographic audits to ensure the acoustic classification mapping performs equitably across diverse regional accents and dialects.

Thank You